Pan-Genome-Guided Reverse Vaccinology and Immunoinformatics Pipeline

A step-by-step computational methodology tutorial for multi-epitope vaccine design against *Providencia stuartii*

GitHub repository documentation draft

This document organizes the methodology reported in the research manuscript into a sequential workflow. It is intended as the foundation for a public GitHub README and reproducibility package.

Scope: genome retrieval, pan-genome analysis, antigen and epitope screening, vaccine construction, structural assessment, immune simulation, docking, molecular dynamics, and in-silico cloning.

Study dataset: 31 complete *P. stuartii* genome assemblies. **Genome retrieval date reported:** 20 March 2026.

Part I. Genome retrieval and antigen discovery

Step 1. Whole-genome sequence retrieval

Complete genome assemblies of P. stuartii were retrieved from NCBI. Thirty-one complete assemblies were selected for comparative genomic and pan-genome analyses. Genome metadata was retrieved using the NCBI Datasets command-line interface.

NCBI Datasets: https://www.ncbi.nlm.nih.gov/datasets/genome/
NCBI Datasets CLI documentation: https://www.ncbi.nlm.nih.gov/datasets/docs/v2/

Parameters / criteria: Assembly level: complete. Retrieval date: 20 March 2026.

Reported output: 31 complete P. stuartii genome assemblies.

Step 2. Genome annotation

Genome assemblies were annotated to identify protein-coding sequences and assign functional annotations. The resulting GFF3 files were used for pan-genome analysis.

Prokka v1.14.6: https://github.com/tseemann/prokka

Step 3. Pan-genome analysis

Annotated GFF3 files were analyzed to identify gene clusters shared among isolates. Conserved core genes were prioritized, and literature-derived outer membrane, flagellar, and virulence-associated genes were added.

Roary v3.11.2: https://github.com/sanger-pathogens/Roary

Parameters / criteria: BLASTp identity: 95%; core: 99–100%; soft core: 95–<99%; shell: 15–<95%; cloud: <15%.

Reported output: 344 core genes; 340 retained after excluding hypothetical or ambiguously annotated genes. Seventeen additional genes yielded 357 initial candidates.

Part II. Antigen and epitope screening

Step 4. Antigenicity prediction

Candidate proteins were evaluated for predicted antigenicity. Proteins exceeding the selected threshold were retained.

VaxiJen: https://www.ddg-pharmfac.net/vaxijen/VaxiJen/VaxiJen.html

Parameters / criteria: Threshold: 0.4. Confirm the exact VaxiJen version used.

Step 5. Subcellular localization prediction

Protein localization was predicted to prioritize extracellular and outer membrane-associated proteins that may be accessible to the host immune system.

PSORTb: https://www.psort.org/psortb/

Step 6. Transmembrane helix prediction

Transmembrane helices were predicted to evaluate membrane topology and structural accessibility. The manuscript reports excluding candidates with more than one predicted transmembrane helix.

TMHMM 2.0: https://services.healthtech.dtu.dk/services/TMHMM-2.0/
CCTOP: https://cctop.ttk.hu/

Step 7. Signal peptide prediction

Signal peptides were predicted to assess protein export and membrane localization. Signal peptide and lipoprotein signal peptide predictions were considered.

SignalP 6.0: https://services.healthtech.dtu.dk/services/SignalP-6.0/

Step 8. Candidate-protein allergenicity and toxicity screening

Candidate proteins were screened for predicted allergenicity and toxicity. Candidates predicted to be allergenic or toxic were excluded.

AllerTOP: https://www.ddg-pharmfac.net/AllerTOP/
ToxinPred 3.0: https://webs.iiitd.edu.in/raghava/toxinpred3/

Step 9. Linear B-cell epitope prediction

Linear B-cell epitopes were predicted from selected candidate proteins to identify peptide regions with potential B-cell recognition.

IEDB B-cell tools: https://tools.iedb.org/bcell/

Parameters / criteria: BepiPred Linear Epitope Prediction 2.0.

Step 10. CTL epitope prediction

MHC class I-restricted CTL epitopes were predicted from candidate proteins. Peptides below the reported percentile-rank threshold were retained.

IEDB Next-Generation Tools: https://nextgen-tools.iedb.org/

Parameters / criteria: Percentile rank <1.

Step 11. HTL epitope prediction

MHC class II-restricted HTL epitopes were predicted using NetMHCIIpan. Peptides below the reported percentile-rank threshold were retained.

IEDB Next-Generation Tools: https://nextgen-tools.iedb.org/

Parameters / criteria: Percentile rank <10. Confirm the species and allele panel; the manuscript refers to mouse MHC-II alleles.

Step 12. IFN-γ induction prediction

Selected HTL epitopes were evaluated for their predicted ability to induce IFN-γ responses. Only positive SVM predictions were retained.

IFNepitope: https://webs.iiitd.edu.in/raghava/ifnepitope/

Parameters / criteria: SVM score >0.

Step 13. Epitope antigenicity, allergenicity, and toxicity screening

Selected B-cell, CTL, and HTL epitopes were evaluated for predicted antigenicity, allergenicity, and toxicity.

VaxiJen: https://www.ddg-pharmfac.net/vaxijen3/home/
AllerTOP: https://www.ddg-pharmfac.net/AllerTOP/
ToxinPred 3.0: https://webs.iiitd.edu.in/raghava/toxinpred3/

Step 14. Population coverage analysis

Population coverage of selected CTL and HTL epitopes was estimated using their MHC-restricting alleles. Individual and combined class I and class II coverage were assessed.

IEDB Population Coverage: https://tools.iedb.org/population/

Part III. Vaccine construction and physicochemical assessment

Step 15. Multi-epitope vaccine construct assembly

Selected B-cell, CTL, and HTL epitopes were assembled into one construct. CTB was incorporated as an adjuvant, and PADRE was included as a helper T-cell epitope.

Parameters / criteria: Linkers: EAAAK (adjuvant fusion), KK (B-cell epitopes), GPGPG (HTL epitopes), AAY (CTL epitopes).

Step 16. Physicochemical property analysis

Molecular weight, theoretical isoelectric point, half-life, instability index, aliphatic index, and GRAVY score were calculated.

ExPASy ProtParam: https://web.expasy.org/protparam/

Step 17. Protein solubility prediction

Predicted solubility following heterologous expression in E. coli was assessed.

SOLpro: https://scratch.proteomics.ics.uci.edu/

Part IV. Structural modeling and refinement

Step 18. Three-dimensional structure prediction

The vaccine construct structure was predicted using AlphaFold 3. Five models were generated, and the top-ranked model was selected using model ranking and confidence measures.

AlphaFold 3: https://alphafoldserver.com/

Parameters / criteria: Reported measures: pLDDT, PAE, pTM, and ranking score.

Step 19. Three-dimensional structure refinement

The selected AlphaFold 3 model was refined to improve structural geometry and model quality.

GalaxyRefine: https://galaxy.seoklab.org/

Step 20. Structural validation

The refined structure was evaluated for stereochemical quality, non-bonded interactions, and overall structural reliability.

SAVES: https://saves.mbi.ucla.edu/
ProSA-web: https://prosa.services.came.sbg.ac.at/prosa.php

Parameters / criteria: PROCHECK and ERRAT were accessed through SAVES, as reported.

Step 21. Conformational B-cell epitope prediction

Discontinuous or conformational B-cell epitopes were predicted using the refined three-dimensional vaccine structure.

IEDB ElliPro: https://tools.iedb.org/ellipro/

Step 22. Disulfide engineering

Potential residue pairs for disulfide bond formation were identified to investigate possible improvements in structural rigidity.

Disulfide by Design 2: http://cptweb.cpt.wayne.edu/DbD2/

Parameters / criteria: Reported criteria: energy threshold 2.2; chi3 angle range –87° to +97°.

Part V. Immune simulation and molecular docking

Step 23. In-silico immune simulation

The predicted immune response following repeated vaccine administration was evaluated. Humoral and cellular responses, antibody production, cytokine levels, and immune-cell populations were examined.

C-ImmSim: https://kraken.iac.rm.cnr.it/C-IMMSIM/

Parameters / criteria: Three administrations; simulation time steps 1, 84, and 168; 1,050 total simulation steps; one step represents approximately 8 hours. Other parameters were reported as default.

Step 24. Molecular docking

Protein–protein docking was performed to investigate the predicted interaction between the vaccine construct and human TLR4. Control structures and an adjuvant-free construct were included.

ClusPro: https://cluspro.org/
RCSB PDB: https://www.rcsb.org/

Parameters / criteria: Receptor: human TLR4, PDB ID 4G8A. Controls reported: Lipid A, PDB ID 1FXI; GFP, PDB ID 1EMA; and vaccine without adjuvant.

Step 25. Docking interaction and binding-affinity analysis

Selected docked complexes were analyzed to characterize intermolecular interactions and estimate binding affinity.

PRODIGY: https://wenmr.science.uu.nl/prodigy/
PDBsum: https://www.ebi.ac.uk/thornton-srv/databases/pdbsum/
BIOVIA Discovery Studio: https://discover.3ds.com/discovery-studio-visualizer-download

Step 26. Normal mode analysis

Normal mode analysis was performed to evaluate intrinsic flexibility, deformability, and collective motions of the selected vaccine–TLR4 complex.

iMODS: https://imods.iqfr.csic.es/

Part VI. Molecular dynamics and trajectory analyses

Step 27. Molecular dynamics simulation

Molecular dynamics simulations were conducted for the unbound vaccine construct and vaccine–TLR4 complex to investigate conformational stability and dynamic behavior over 100 ns.

GROMACS: https://www.gromacs.org/
GROMACS 2025.3 documentation: https://manual.gromacs.org/documentation/2025.3/

Parameters / criteria: GROMOS54a7; TIP3P water; 15 Å boundary; 0.15 M NaCl; 5,000 steepest-descent minimization steps; 100 ps NVT and 100 ps NPT; 300 K; 1 bar; LINCS; PME; 9.0 Å electrostatic cutoff; trajectory interval 1 ps; production run 100 ns.

Reported output: RMSD, RMSF, radius of gyration (Rg), and hydrogen-bond analyses.

Step 28. Binding free-energy calculation

MM-PBSA analysis was performed using the final molecular dynamics trajectories to estimate binding free energy of the vaccine–TLR4 complex.

g_mmpbsa: https://github.com/rjdkmr/g_mmpbsa

Parameters / criteria: Reported terms: van der Waals (ΔVDW), electrostatic (ΔELE), polar solvation (ΔGB), and non-polar solvation (ΔSA).

Step 29. Principal component analysis

PCA was conducted to investigate dominant collective motions and conformational dynamics during the molecular dynamics simulation.

GROMACS documentation: https://manual.gromacs.org/documentation/2025.3/

Parameters / criteria: The manuscript reports the GROMACS covariance-analysis workflow (gmxcovar); retain exact commands and scripts.

Step 30. Free energy landscape analysis

Free energy landscapes were generated from the probability distributions of the first two principal components to examine conformational states and energy minima.

GROMACS documentation: https://manual.gromacs.org/documentation/2025.3/

Step 31. Dynamic cross-correlation matrix analysis

Dynamic cross-correlation analysis was used to investigate correlated and anti-correlated residue movements. Trajectories were converted from XTC to DCD format before analysis.

VMD: https://www.ks.uiuc.edu/Research/vmd/
Bio3D: https://thegrantlab.org/bio3d/

Part VII. Codon optimization and in-silico cloning

Step 32. Reverse translation and codon optimization

The final vaccine amino-acid sequence was reverse-translated into a nucleotide sequence. Codon optimization was performed for the intended E. coli expression host.

EMBOSS Backtranseq: https://www.ebi.ac.uk/Tools/st/emboss_backtranseq/
JCat: https://www.jcat.de/

Parameters / criteria: Evaluation measures: codon adaptation index (CAI) and GC content.

Step 33. In-silico cloning

The optimized nucleotide sequence was inserted into the pET28a(+) expression vector using SnapGene. Restriction sites were incorporated to support the cloning design.

SnapGene: https://www.snapgene.com/

Parameters / criteria: Expression vector: pET28a(+). Restriction enzymes: NdeI and XhoI.

Computational workflow summary

1 Genome retrieval and selection
2 Genome annotation
3 Pan-genome analysis
4 Antigenicity, localization, topology, and signal-peptide screening
5 Protein allergenicity and toxicity screening
6 B-cell and T-cell epitope prediction
7 IFN-γ prediction and epitope safety screening
8 Population coverage analysis
9 Multi-epitope vaccine construction
10 Physicochemical and solubility analysis
11 3D structure prediction, refinement, and validation
12 Conformational B-cell epitope prediction
13 Disulfide engineering
14 Immune simulation
15 Molecular docking and interaction analysis
16 Normal mode analysis
17 Molecular dynamics simulation
18 Binding free-energy analysis
19 PCA, FEL, and DCCM
20 Reverse translation, codon optimization, and in-silico cloning

Scope and interpretation

This is a computational vaccine-design and evaluation pipeline. Antigenicity, allergenicity, toxicity, immunogenicity, receptor binding, population coverage, and structural stability are predictions or model-derived estimates. They do not establish experimental immunogenicity, safety, expression, or protective efficacy. Experimental validation is required.

Suggested article citation: Tanvir Ahmed Sujon et al. Integrated pan-genomics and immunoinformatics for the design of a multi-epitope vaccine candidate against Providencia stuartii. Add the final journal, year, DOI, and author-approved citation details when available.

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